Lab 9 Contribution Journal

TheBetaThetas | ITAI 1371 | Lab 9: Ensemble Methods

Brandy Gonzales

For Lab 9, I created the shared Google Colab notebook where our team collects each member's reflection and contribution entries. I worked through the Ensemble Methods lab independently, wrote the missing code, and fixed my errors (a typo, a capital-letter mix-up between x_train and X_train, and a skipped cell) until the Decision Tree and Random Forest both ran correctly. I also answered all three knowledge check questions.

Marlon Roberts

What I contributed to in this lab is that I made the repository open for my team to use. I completed the module and learned a lot from the ensemble method. I implemented the random forest ensemble method, extracted and rendered the feature importance. I pulled the underlying structural weights using random_forest.feature_importances_, calculated an ascending array sort index using np.argsort(), and successfully generated a clear, horizontal bar plot complete with proper figure dimensions, matching target tick-labels (iris.feature_names), and axis labels to interpret the model's inner decision-making. Doing this lab, we were able to see the importance of these calculations.

Derrick K. Gibbs-McGlaston

I prepared the training and testing data, trained a Decision Tree, and evaluated its test accuracy. I then trained a Random Forest using 100 trees and compared the results using the same test data. I generated a feature importance chart and used it to identify the flower measurements that contributed most to the Random Forest predictions. I also answered the knowledge-check questions and reviewed how the results relate to the purpose of ensemble learning. I checked that the notebook cells ran and that the printed results and chart appeared.

Victor Djuma

My individual contribution to Lab 09 focused on completing, running, checking, and understanding the machine learning notebook for Ensemble Methods. I worked through each required section of the lab and made sure the code executed correctly before moving to the next part.

I started by preparing the Iris dataset using the setup code provided in the notebook. The dataset was divided into training and testing data so the models could be trained on one portion and evaluated on another portion. I reviewed the setup carefully because the rest of the lab depended on having the training and testing data prepared correctly.

For the first modeling task, I completed the code for the Decision Tree classifier. I created the Decision Tree using a fixed random state, trained it using the training data, made predictions on the test data, and calculated the accuracy. The Decision Tree achieved 100% accuracy. I checked the output to make sure the model had trained and evaluated successfully.

For the second modeling task, I completed the Random Forest classifier. I created a Random Forest with 100 trees and trained it using the same training data. I then made predictions on the test set and calculated its accuracy. The Random Forest also achieved 100% accuracy. I compared this result with the single Decision Tree and noted that both models produced the same accuracy in this lab. This comparison helped me better understand that even though Random Forest is usually more stable, it does not always have to produce a higher accuracy score on every dataset.

I also completed the feature importance section of the notebook. I used the trained Random Forest model to extract the importance of the four Iris features and created a horizontal bar graph to visualize the results. From this graph, I identified petal width and petal length as the two most important features for classifying the flowers.

Another part of my contribution was completing the Knowledge Check questions. I explained the main idea behind ensemble methods, compared the Decision Tree and Random Forest results, and interpreted the feature importance graph. I tried to answer these questions in my own words so that the answers reflected my understanding of the lab rather than simply repeating definitions.

I also made sure the notebook was organized correctly, that the code cells were executed, that the outputs were visible, and that the feature importance graph appeared correctly. After completing the notebook, I reviewed the final results and prepared the notebook for PDF submission.

Through my contribution to this lab, I gained practical experience with Decision Trees, Random Forests, ensemble methods, model evaluation, and feature importance. My main responsibility was completing the coding tasks, verifying the outputs, interpreting the results, and making sure the final notebook clearly demonstrated the required workflow.

Pablo Lara

Not yet received.